Code JupyterLab Python [conda env:base] * Anaconda Toolbox

[1]:
```
import numpy as np
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
```

[2]:
```
(x_train, y_train), (x_test, y_test) = keras.datasets.mnist.load_data()
x_train = x_train.astype("float32") / 255.0
x_test = x_test.astype("float32") / 255.0
```

[3]:
```
x_train = x_train[y_train == 0]
x_test = x_test[np.isin(y_test, [0, 1])]
y_test = y_test[np.isin(y_test, [0, 1])]
```

[4]:
```
x_train = x_train.reshape(-1, 784)
x_test = x_test.reshape(-1, 784)
```

[5]:
```
input_layer = keras.Input(shape=(784,))
encoded = layers.Dense(128, activation="relu")(input_layer)
encoded = layers.Dense(32, activation="relu")(encoded)

decoded = layers.Dense(128, activation="relu")(encoded)
output_layer = layers.Dense(784, activation="sigmoid")(decoded)

autoencoder = keras.Model(input_layer, output_layer)
```

[6]:
```
autoencoder.compile(
```

Code JupyterLab Python [conda env:base] * Anaconda Toolbox

[6]:
```
autoencoder.compile(
    optimizer="adam",
    loss="mse",
    metrics=["mae"]
)
```

```
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be us
ed. Please use WSL2 or the TensorFlow-DirectML plugin.
```

[7]:
```
autoencoder.fit(
    x_train,
    x_train,
    epochs=5,
    batch_size=128,
    validation_split=0.1
)
```

```
Epoch 1/5
42/42 ━━━━━━━━━━━━━━━━━━━━ 6s 48ms/step - loss: 0.1069 - mae: 0.2333 - val_loss: 0.0578 - val_mae: 0.1257
Epoch 2/5
42/42 ━━━━━━━━━━━━━━━━━━━━ 1s 14ms/step - loss: 0.0489 - mae: 0.1108 - val_loss: 0.0413 - val_mae: 0.0967
Epoch 3/5
42/42 ━━━━━━━━━━━━━━━━━━━━ 1s 13ms/step - loss: 0.0393 - mae: 0.0959 - val_loss: 0.0347 - val_mae: 0.0881
Epoch 4/5
42/42 ━━━━━━━━━━━━━━━━━━━━ 1s 13ms/step - loss: 0.0333 - mae: 0.0860 - val_loss: 0.0298 - val_mae: 0.0789
Epoch 5/5
42/42 ━━━━━━━━━━━━━━━━━━━━ 1s 22ms/step - loss: 0.0288 - mae: 0.0772 - val_loss: 0.0260 - val_mae: 0.0714
```

[7]:
```
<keras.src.callbacks.history.History at 0x1fb67ab4590>
```

[9]:
```
reconstructed = autoencoder.predict(x_test)
```

```
    x_train,
    epochs=5,
    batch_size=128,
    validation_split=0.1
)
```

```
Epoch 1/5
42/42 ━━━━━━━━━━━━━━━━━━━━ 6s 48ms/step - loss: 0.1069 - mae: 0.2333 - val_loss: 0.0578 - val_mae: 0.1257
Epoch 2/5
42/42 ━━━━━━━━━━━━━━━━━━━━ 1s 14ms/step - loss: 0.0489 - mae: 0.1108 - val_loss: 0.0413 - val_mae: 0.0967
Epoch 3/5
42/42 ━━━━━━━━━━━━━━━━━━━━ 1s 13ms/step - loss: 0.0393 - mae: 0.0959 - val_loss: 0.0347 - val_mae: 0.0881
Epoch 4/5
42/42 ━━━━━━━━━━━━━━━━━━━━ 1s 13ms/step - loss: 0.0333 - mae: 0.0860 - val_loss: 0.0298 - val_mae: 0.0789
Epoch 5/5
42/42 ━━━━━━━━━━━━━━━━━━━━ 1s 22ms/step - loss: 0.0288 - mae: 0.0772 - val_loss: 0.0260 - val_mae: 0.0714
```

[7]:
```
<keras.src.callbacks.history.History at 0x1fb67ab4590>
```

[9]:
```
reconstructed = autoencoder.predict(x_test)

error = np.mean((x_test - reconstructed) ** 2, axis=1)

threshold = np.percentile(error[y_test == 0], 95)

print("Anomaly threshold:", threshold)
```

```
67/67 ━━━━━━━━━━━━━━━━━━━━ 0s 3ms/step
Anomaly threshold: 0.04709582
```

[ ]: